

Project: EquipTrack – Industrial Equipment Tracking & Maintenance System

1. Introduction

This document outlines the Low-Level Design (LLD) for **EquipTrack**, an industrial equipment tracking and maintenance system designed for manufacturing plants. The system digitizes equipment registration, automates maintenance scheduling, monitors performance metrics, and provides analytics for downtime reduction and operational efficiency.

Supports backend development using **Java (Spring Boot)** and **.NET (ASP.NET Core)**.

2. Module Overview

- 2.1 Equipment Registration & Lifecycle Module
- 2.2 Maintenance Scheduling & Execution Module
- 2.3 Performance Monitoring & Alerting Module
- 2.4 Spare Parts & Inventory Management Module
- 2.5 Downtime Analytics & Reporting Module

3. Architecture Overview

3.1 Architectural Style

- **Frontend:** Angular/React for operator and admin dashboards
- **Backend:** REST APIs for equipment tracking and maintenance workflows
- **Database:** MySQL/SQL Server for structured equipment and maintenance data

3.2 Component Interaction

- Frontend captures equipment details and maintenance logs
- Backend processes schedules, alerts, and analytics

4. Module-Wise Design

4.1 Equipment Registration & Lifecycle Module

4.1.1 Features

- Register new equipment with specifications and vendor details
- Track lifecycle stages: Installed, Operational, Under Maintenance, Decommissioned

4.1.2 Entities

- **Equipment**
 - EquipmentID
 - Name
 - Category
 - Vendor
 - InstallationDate
 - Status (INSTALLED, OPERATIONAL, MAINTENANCE, DECOMMISSIONED)

4.2 Maintenance Scheduling & Execution Module

4.2.1 Features

- Schedule preventive and corrective maintenance
- Log maintenance activities and technician notes

4.2.2 Entities

- **MaintenanceLog**
 - MaintenanceID
 - EquipmentID
 - MaintenanceType (PREVENTIVE, CORRECTIVE)
 - ScheduledDate
 - CompletionDate
 - Technician
 - Status (SCHEDULED, COMPLETED)

4.3 Performance Monitoring & Alerting Module

4.3.1 Features

- Monitor metrics like temperature, vibration, and load
- Trigger alerts for threshold breaches

4.3.2 Entities

- **PerformanceMetric**
 - MetricID
 - EquipmentID
 - MetricType (TEMPERATURE, VIBRATION, LOAD)
 - Value
 - RecordedDate
 - AlertStatus (NORMAL, WARNING, CRITICAL)

4.4 Spare Parts & Inventory Management Module

4.4.1 Features

- Track spare parts usage and stock levels
- Generate reorder alerts

4.4.2 Entities

- **SparePart**
 - PartID
 - Name
 - EquipmentID
 - QuantityAvailable
 - ReorderLevel

4.5 Downtime Analytics & Reporting Module

4.5.1 Features

- Analyze downtime patterns and root causes
- Generate reports for operational efficiency

4.5.2 Entities

- **DowntimeReport**
 - ReportID
 - EquipmentID
 - DowntimeDuration
 - Cause
 - GeneratedDate
 - Recommendations

5. Deployment Strategy

5.1 Local Deployment

- Developer machines with sample equipment and maintenance data

5.2 Staging and Production Environments

- Cloud deployment with secure access and monitoring dashboards

6. Database Design

6.1 Tables and Relationships

- Equipment → MaintenanceLog → PerformanceMetric → SparePart → DowntimeReport

7. User Interface Design

7.1 Wireframes

- **Operator Portal:** Log maintenance, view equipment status
- **Admin Dashboard:** Schedule tasks, monitor alerts
- **Analytics Console:** View downtime trends and performance metrics

8. Non-Functional Requirements

8.1 Performance

- Handle 10,000 equipment metric logs per hour

8.2 Usability

- Intuitive UI for factory operators and engineers
- Inline help and tooltips for maintenance workflows

8.3 Security

- Role-based access and encrypted data transmission
- Compliance with ISO 27001 and industry safety standards

8.4 Scalability

- Support for multiple plants and equipment categories

9. Assumptions and Constraints

9.1 Assumptions

- Equipment vendors provide digital specifications
- Maintenance logs are entered via web or mobile portal

9.2 Constraints

- Initial rollout for mechanical equipment only